Pandas 3.0.6 Quick Revision & Reference Guide

Library Version: 3.0.6 (Stable) | **Release Date:** September 17, 2026 | **License:** BSD Open Source

1. Library Overview & Architecture

pandas is an open-source, BSD-licensed library providing high-performance, easy-to-use data structures and data analysis tools for the Python programming language.

> **Documentation Source Context:** Grounded in the official Pandas 3.0.6 documentation landing page. The documentation is structured into five primary sections: Getting Started, User Guide, API Reference, Developer Guide, and Release Notes (What's New).

Documentation Module	Core Focus	Target Audience / Use Case
Getting Started	Introductory tutorials, installation instructions, and core concepts overview.	Beginners & new users
User Guide	In-depth background explanation of key concepts and data analysis workflows.	Intermediate users & practitioners
API Reference	Detailed method specifications, signature parameters, and return types.	Developers & reference lookups
Developer Guide	Guidelines for contributing, bug reporting, documentation typos, and codebase improvements.	Open-source contributors
Release Notes	Summary of changes, new features, deprecations, and performance updates in pandas 3.0.	Upgrading developers

2. Core Data Structures & Basic Operations (Code Reference)

The API Reference and User Guide cover pandas' two fundamental data structures: **Series** (1D labeled array) and **DataFrame** (2D labeled tabular structure). Below are revision code templates.

A. Creating Series and DataFrames

```
import pandas as pd
import numpy as np

# 1. Creating a 1D Series
s = pd.Series([10, 20, 30, 40], index=['a', 'b', 'c', 'd'], name="Metrics")

# 2. Creating a 2D DataFrame from a dictionary
data = {
    'Department': ['Sales', 'Engineering', 'Marketing', 'Sales'],
    'Employees': [12, 24, 8, 15],
    'Budget_kUSD': [150.5, 320.0, 95.4, 180.2]
}
df = pd.DataFrame(data)
print(df.head())
```

B. Data Inspection & Summary Statistics

```
# Quick structural inspection
print(df.info())         # Data types and non-null counts
print(df.describe())     # Summary statistics for numeric columns
print(df.shape)          # (rows, columns) shape tuple
print(df.columns)        # Column labels Index
```

C. Selection & Filtering (loc vs iloc)

```
# Label-based indexing (.loc) vs Position-based indexing (.iloc)
sales_dept = df.loc[df['Department'] == 'Sales', ['Department', 'Budget_kUSD']]

# First 2 rows, first 2 columns
sub_table = df.iloc[0:2, 0:2]
```

3. Common Data Manipulation Workflows

```
# Grouping & Aggregation
group_summary = df.groupby('Department').agg(
    Total_Employees=('Employees', 'sum'),
    Avg_Budget=('Budget_kUSD', 'mean')
).reset_index()

# Handling Missing Values
df_clean = df.dropna(subset=['Budget_kUSD'])  # drop missing values
df_filled = df.fillna({'Budget_kUSD': 0})      # fill missing values

# Reading & Writing Data
# df = pd.read_csv('data.csv')
# df.to_csv('output.csv', index=False)
```

4. Documentation Links & Official Resources

Resource Title	URL / Access Point	Purpose
Zipped HTML Documentation	https://pandas.pydata.org/docs/pandas.zip	Offline documentation package
PyPI Binary Installers	https://pypi.org/project/pandas	pip install package repository
GitHub Source Repository	https://github.com/pandas-dev/pandas	Source code & issue tracking
Stack Overflow Support	https://stackoverflow.com/questions/tagged/pandas	Community Q&A; support forum